

WATER SPLITTING

A highly active and stable $\text{IrO}_x/\text{SrIrO}_3$ catalyst for the oxygen evolution reaction

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Oxygen electrochemistry plays a key role in renewable energy technologies such as fuel cells and electrolyzers, but the slow kinetics of the oxygen evolution reaction (OER) limit the performance and commercialization of such devices. Here we report an iridium oxide/strontium iridium oxide ($\text{IrO}_x/\text{SrIrO}_3$) catalyst formed during electrochemical testing by strontium leaching from surface layers of thin films of SrIrO_3 . This catalyst has demonstrated specific activity at 10 milliamperes per square centimeter of oxide catalyst (OER current normalized to catalyst surface area), with only 270 to 290 millivolts of overpotential for 30 hours of continuous testing in acidic electrolyte. Density functional theory calculations suggest the formation of highly active surface layers during strontium leaching with IrO_3 or anatase IrO_2 motifs. The $\text{IrO}_x/\text{SrIrO}_3$ catalyst outperforms known IrO_x and ruthenium oxide (RuO_x) systems, the only other OER catalysts that have reasonable activity in acidic electrolyte.

Advances in renewable energy largely rely on technologies for energy conversion and storage to mitigate intermittency issues associated with wind and sunlight. Electrochemical processes such as water splitting and CO_2 reduction have been widely studied for conversion of electrical energy into fuels and chemicals. Water oxidation, or the oxygen evolution reaction (OER), plays a key role by providing the protons and electrons needed for these processes; improving the efficiency and durability of OER catalysts has a direct impact on device efficiency and cost effectiveness. Here we report an OER catalyst that demonstrates activity and stability in acid electrolyte.

Decades of research investigating catalysts for the OER have led to substantial advancements in the field and the development of active catalysts. However, even the best currently known catalysts require >320 mV overpotential to reach $10 \text{ mA}/\text{cm}^2_{\text{oxide}}$ (*1–8*), a measure of intrinsic activity acquired by normalizing the OER current to the catalyst surface area. To partially overcome the low intrinsic activity of known OER catalysts, the activity of an OER electrode can be improved on a geometric-area basis by increasing catalyst loadings or by improving catalyst morphologies to expose large surface areas. Such approaches, however, can ultimately become

limited by mass transport and/or catalyst conductivity. Another major concern is the cost and scarcity of precious metal-based OER catalysts, particularly in acidic electrolyte.

Whereas there are a plethora of precious and nonprecious metal catalysts that exhibit activity and stability in alkaline electrolyte, IrO_x and RuO_x are currently the only known materials that can reach $\sim 5 \text{ mA}/\text{cm}^2_{\text{oxide}}$ with overpotentials less than 750 mV in acidic electrolyte (*1, 9*). New OER catalysts for use in acidic environments could have a direct impact on improving polymer electrolyte membrane-based water electrolyzers, a recent technology that overcomes many of the disadvantages of conventional alkaline electrolyzers (*10*) but in which OER catalysts are primarily restricted to IrO_x .

The recent development of techniques for high-quality growth of epitaxial complex oxide thin films and heterostructures provides an intriguing opportunity for the conception and fundamental study of OER catalysts (*11, 12*). Using such an approach, we developed a catalyst $\text{IrO}_x/\text{SrIrO}_3$, which is stable in acidic electrolyte and substantially outperforms known IrO_x catalysts. We applied density functional theory (DFT) calculations and several materials characterization techniques to understand the possible active sites contributing to the extremely high activity.

The bulk material of this catalyst consists of epitaxial SrIrO_3 thin films deposited on SrTiO_3 substrates by pulsed laser deposition (PLD; details are provided in the supplementary materials) (*12*). The catalyst activity and stability was assessed in $0.5 \text{ M H}_2\text{SO}_4$ by using alternating cyclic voltammetry and chronopotentiometric holds at $10 \text{ mA}/\text{cm}^2_{\text{geo}}$ (current normalized to the geometric surface area of the electrode or the catalyst substrate) in a procedure similar to that which

has been used in benchmarking studies for other electrochemical catalysts (*1*). Figure 1A shows the potential required to reach $10 \text{ mA}/\text{cm}^2_{\text{geo}}$ during a 30-hour stability test, and fig. S1A shows example cyclic voltammograms (CVs) taken periodically during testing. Figure 1B shows a Tafel plot of the intrinsic, or specific, activities for both films over the course of the stability test, compared with the best reported catalysts in acidic electrolyte (*1, 13*). Initially, the films require 340 mV of overpotential (measured by CV) to reach $10 \text{ mA}/\text{cm}^2_{\text{oxide}}$ and improve to 320 mV of overpotential after 10 min of testing. The catalyst continues to increase in activity during the first 2-hour chronopotentiometric hold and requires only 270 to 290 mV of overpotential to reach $10 \text{ mA}/\text{cm}^2_{\text{oxide}}$ for the remainder of the measurements. Figure 1B also presents specific activity for a flat 40-nm epitaxial rutile IrO_2 (110) film deposited on TiO_2 (110) by PLD as a direct comparison with our sample morphology. This rutile IrO_2 film has similar activity to rutile IrO_2 and RuO_2 nanoparticles (*13*) when normalized to the catalyst surface area, but the $\text{IrO}_x/\text{SrIrO}_3$ catalyst outperforms all of these by two orders of magnitude. Whereas Fig. 1B compares the $\text{IrO}_x/\text{SrIrO}_3$ catalyst with some of the best catalysts showing activity and reasonable stability in acidic electrolyte, for further comparison, another Tafel plot is shown in fig. S1B that includes the best reported catalysts in both acidic and alkaline electrolyte.

DFT calculations were performed to provide further understanding of the nature of the SrIrO_3 surface and its high activity. Consistent with previous studies (*14, 15*), we assumed a four-step OER mechanism that proceeds through OH^* , O^* , and OOH^* (where the asterisk indicates surface-bound species), and we evaluated OER activity on the basis of the theoretical overpotential, which is the overpotential required to make all elementary steps exergonic (further model details are included in the supplementary materials).

DFT simulations of the IrO_2 termination of SrIrO_3 (001) (Fig. 2B) yielded a theoretical overpotential of 0.59 V, which is well above the value that we calculated for rutile IrO_2 (110) (0.38 V). This suggests that surface sites at the (001) facet may not be responsible for the high activity of SrIrO_3 that we observed experimentally. It is well known that alkaline earth metals may leach from perovskites under aqueous conditions (*16*), and in the case of SrIrO_3 , we calculated a thermodynamic driving force of 3.2 eV per Sr atom to form aqueous Sr^{2+} and rutile IrO_2 at the relevant experimental conditions. We investigated possible motifs of the Sr-deficient $\text{IrO}_x/\text{SrIrO}_3$ surface by examining a variety of overlayer structures that may arise during OER and contribute to catalytic activity.

The first overlayer structure that we considered was anatase IrO_2 . Although rutile is known to be the most stable bulk polymorph of IrO_x , the anatase structure is much more commensurate with the SrIrO_3 perovskite lattice because their calculated lattice constants are within 2% of each other. The anatase and perovskite structures are also very similar when viewed down the c axis—both consist of IrO_6 octahedra that share corners

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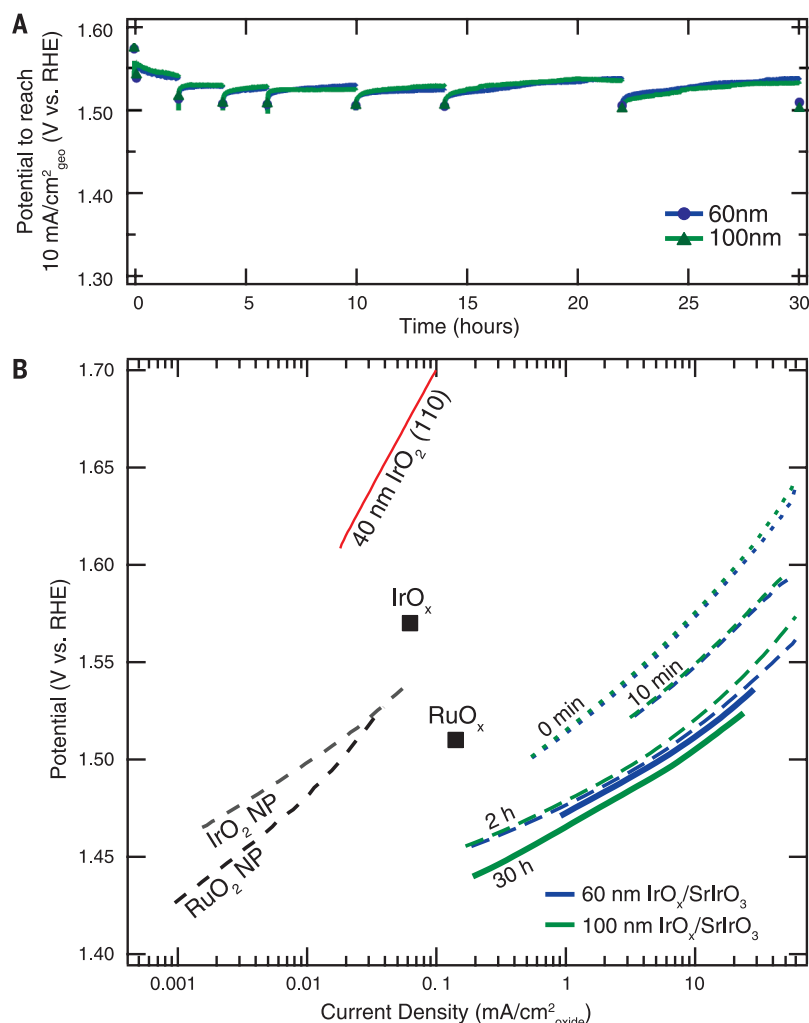


Fig. 1. Electrochemical activity. (A) Potential required to reach 10 mA/cm²_{geo} for IrO_x/SrIrO₃ films over 30 hours of electrochemical testing, measured using cyclic voltammetry (markers) and chronopotentiometric holds (lines). (B) Tafel plot comparing specific activity of IrO_x/SrIrO₃ with that of OER catalysts in acidic electrolyte. Green lines show 100-nm IrO_x/SrIrO₃ and blue lines show 60-nm IrO_x/SrIrO₃ specific activity at various points during electrochemical stability testing. The red line shows specific activity for a 40-nm rutile IrO₂ (110) film. Gray and black lines and squares are duplicated from literature (1, 13). NP, nanoparticles. RHE, reversible hydrogen electrode.

in the (001) planes—suggesting that a transformation between them may be kinetically accessible. Furthermore, the growth of micron-sized (001)-oriented TiO₂ anatase crystallites has been observed experimentally after soaking SrTiO₃ in water (17). On replacement of the top two layers of SrIrO₃ with anatase IrO₂ (001) (Fig. 2C), we calculated a stabilization of 2.2 eV per Sr atom and a theoretical overpotential of 0.30 V. For the limiting case of pure anatase IrO₂ (001) (Fig. 2D), which may occur in thicker overlayers where strain has been released, we calculated a similar theoretical overpotential of 0.28 V, which is very near the peak of the OER volcano (Fig. 2A). This is in agreement with a recent theoretical study of several metastable polymorphs of IrO₂ in the low coverage limit that found anatase to be the most active (18).

The second IrO_x overlayer structure that we considered was produced by simulated annealing using ab initio (DFT) molecular dynamics with

the IrO₂-anatase/SrIrO₃ structure as an initial guess. This structure can be viewed as similar to ReO₃ with a “sheet” of oxygen removed in the (100) plane, which results in a series of square coordinated IrO₄ planes anchored by SrIrO₃ (001) (Fig. 2E). The stability of this structure is similar to that of the anatase overlayer starting point, but its calculated theoretical overpotential (0.78 V) was found to be much worse. Filling the vacant (100) planes with oxygen to form an IrO₃ film stabilizes the structure by −0.03 eV per O atom at OER equilibrium and improves the theoretical overpotential to 0.47 V (Fig. 2F). Removal of a third layer of Sr further reduces the overpotential value to 0.28 V (Fig. 2G). The observation of nominal Ir oxidation states greater than 4+ is unusual in solid oxides, but we predict their stability under the oxidizing experimental conditions.

Ultimately, our calculations suggest that the activity of SrIrO₃ is improved as Sr leaches into

the electrolyte and that the formation of sites similar to IrO₃ or anatase IrO₂ may be responsible for the experimentally observed activity. Although we cannot exclude other, even more stable and active overlayer structures, these calculations illustrate that Sr deficiencies in the near-surface of SrIrO₃ can induce substantially improved OER activity on Ir sites. It is also worth noting that the surfaces considered in this study are concentrated at the more favorable, lower bound of the previously reported universal scaling relationship between OH and OOH (10) (Fig. 2J). This results in an elevated volcano peak in Fig. 2A and suggests that the most active surfaces considered here are at the limit of what is possible without breaking the scaling between OH and OOH.

After identifying active sites for this catalyst surface that may contribute to the observed activity, materials characterization provided further insight into the catalyst morphology and composition before and after electrochemical testing. Atomic force microscopy (AFM) was used to assess the surface morphology and roughness of these films. Figure 3A presents 2- by 2-μm maps and 2-μm line scans for a 100-nm SrIrO₃ film before and after 30 hours of testing. Although there was some surface rearrangement, the films were notably flat as prepared and exhibited uniform small features after stability testing. The root mean square roughness and surface roughness factors (film surface area normalized to geometric area) were quantified for both the SrIrO₃ and the IrO₂ PLD films and are presented in the table in Fig. 3B (AFM maps and line scans are shown in fig. S3). These measurements were used to normalize the electrochemical activity to the actual catalyst surface area, and the flat nature of the films allows for accurate quantification of specific activity.

The expected bulk crystal structure of SrIrO₃ was confirmed by x-ray diffraction (XRD). It has been previously reported that the perovskite-type SrIrO₃ with orthorhombic symmetry is stabilized on SrTiO₃ (100) (12, 19, 20); therefore, all XRD patterns are labeled using the pseudo-cubic notation for SrIrO₃. Figure 3C shows diffractograms of a 100-nm SrIrO₃ film before and after 30 hours of testing. Both display the characteristic sharp peaks of the SrTiO₃ substrate and the broader peaks of orthorhombic SrIrO₃. There is some observable decrease in SrIrO₃ peak intensities measured after OER testing, which is most visible in the SrIrO₃ (200) peaks (enlarged plot in fig. S4), but the bulk of the film appears to have remained crystalline. XRD results for the IrO₂ PLD film are shown in fig. S5.

The surface composition and chemical state of the IrO_x/SrIrO₃ catalysts were assessed by x-ray photoelectron spectroscopy (XPS). Sample Ir4f and Sr3d peaks are shown in Fig. 3D for a 100-nm IrO_x/SrIrO₃ sample before and after 30 min of electrochemical testing. The Ir4f peak can be fit with two doublets, as has been observed previously (21–23), which likely correspond to screened and unscreened components of an Ir⁴⁺ state. The Sr3d peak can be fit with two doublets before testing,

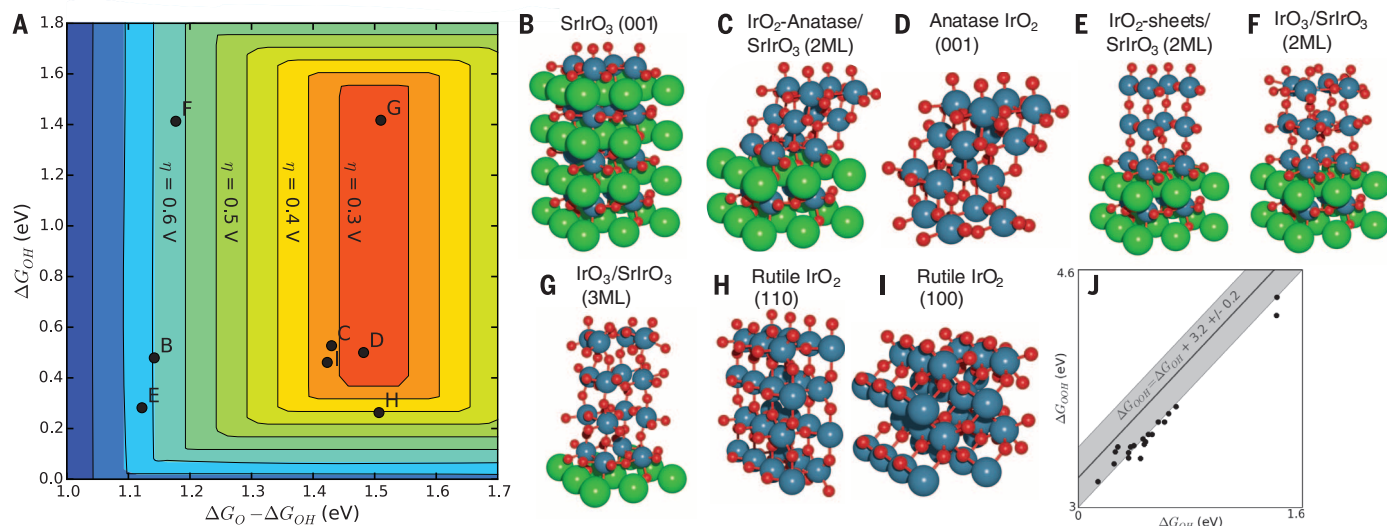


Fig. 2. OER theoretical analysis. (A) Theoretical overpotential (η) volcano plot with O* and OH* binding energies as descriptors, using the scaling relationship between OH and OOH shown in fig. S2B. (B to I) Visual representation of IrO_x and SrIrO₃ surfaces used for the DFT calculations, corresponding to the labeled points in (A) (strontium, green; iridium, blue; oxygen, red). (J) OH* and OOH* binding energies (black circles) overlaid on the universal scaling relationship (gray line with shaded uncertainty) that has been found for many transition metals and transition metal oxides (10). G, Gibbs free energy. ML, monolayers.

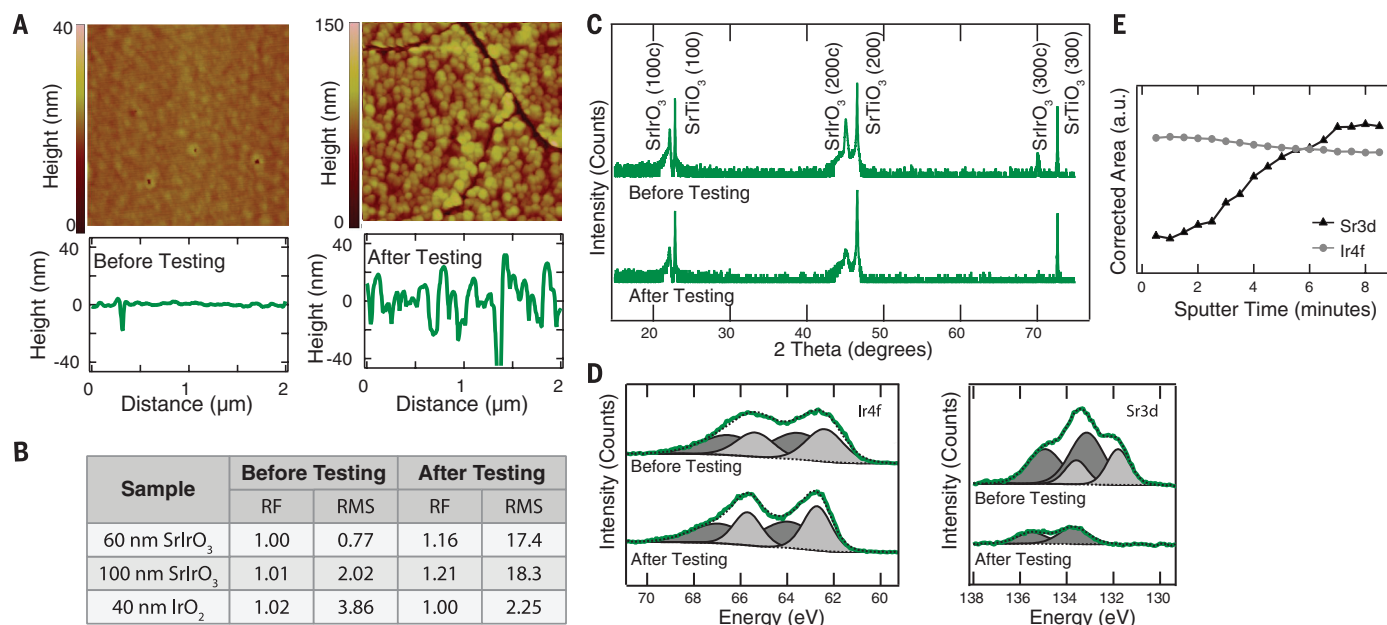


Fig. 3. Material characterization. (A) AFM maps and line scans for a 100-nm SrIrO₃ film before and after 30 hours of OER testing. (B) Table summarizing the roughness factor (RF) and root mean square (RMS) roughness (in nanometers) for two SrIrO₃ samples and one IrO₂ sample before and after 30 hours of OER testing. (C) XRD of a 100-nm SrIrO₃ film before and after 30 hours of OER testing. (D) XPS on the Ir4f and Sr3d peaks for a 100-nm SrIrO₃ film before and after 30 min of OER testing. (E) Depth sputter profile indicating the relative quantities of each element near the surface after 30 min of OER testing. a.u., arbitrary units.

but as Sr is leached during the first 30 min of electrochemical testing, the signal decreases considerably and the spectral shape appears to change; the spectra are only fit with one doublet after testing. Additional XPS spectra are shown for Ir4f and Sr3d peaks in fig. S6, along with a more thorough peak-fitting analysis. A comparison of XPS spectra measured after varying periods of electrochemical testing on sister samples shows the Sr signal decreasing to 25 to 28% of its original value after 30 min and decreasing to about 20% of its original value after 24 hours, during

which time the Ir signal remains constant. This indicates that Sr is leaching from the surface of the catalyst during electrochemical testing and suggests that the rate of Sr leaching is fastest during initial testing and slows or perhaps even stops with additional testing, likely because of kinetic barriers of Sr transport through the lattice. This is in agreement with previous measurements of Sr leaching from SrTiO₃ (16).

Furthermore, a decrease in the Sr XPS signal to 25% of its original value after 30 min of electrochemical testing would indicate that about seven

monolayers of SrIrO₃ at the surface have been completely depleted of Sr. In reality, there is probably a distribution of Sr leaching from the top-most SrIrO₃ layers, as indicated by XPS results in Fig. 3E, which shows the Sr and Ir content as a function of depth from the catalyst surface. Inductively coupled plasma optical emission spectroscopy measurements of the electrolyte after testing provided complementary results, indicating that ~30 to 50% of the Sr is leached from the films in the first 30 min of testing (additional details are provided in the supplementary materials).

Evidence of Sr leaching and minor surface rearrangement of the resulting IrO_x films suggests that an $\text{IrO}_x/\text{SrIrO}_3$ catalyst is formed in situ and becomes increasingly active over the first 2 hours of electrochemical testing. Although the exact structure of the surface formed during reaction is still unknown, DFT calculations have identified possible structures that could be kinetically or epitaxially stabilized by the interface with SrIrO_3 and could contribute to the very high observed activity. The stability and high intrinsic activity of this catalyst show promise for its integration into renewable energy technologies, such as polymer electrolyte membrane electrolyzers, for the sustainable production of hydrogen or other fuels and chemicals.

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ACKNOWLEDGMENTS

This work was supported by the U.S. Department of Energy (DOE), Office of Science, Office of Basic Energy Sciences, under contract no. DE-SC0008685 and through the SUNCAT Center for Interface Science and Catalysis. Partial support was provided by the Center on Nanostructuring for Efficient Energy Conversion at Stanford University, an Energy Frontier Research Center funded by the DOE, Office of Science, under award no. DE-SC0001060. L.C.S. received fellowship support from the DARE (Diversifying Academia, Recruiting Excellence) Doctoral Fellowship supported by the Vice Provost for Graduate Education at Stanford University. Part of this work was performed at the Stanford Nano Shared Facilities. K.N., Y.H., and H.Y.H. acknowledge support from the DOE, Office of Science, Office of Basic Energy Sciences, Division of Materials Sciences and Engineering, under contract no. DE-AC02-76SF00515 (thin film synthesis), and from the Global Climate and Energy Project at Stanford University (modifications under ionic tuning). C.F.D. and C.K. acknowledge support from the NSF Graduate Research Fellowship program under grant no. DGE-114747. C.K. also acknowledges support from the Morgridge Family Stanford Graduate Fellowship.

SUPPLEMENTARY MATERIALS

www.sciencemag.org/content/353/6303/1011/suppl/DC1
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22 February 2016; accepted 9 August 2016
10.1126/science.aaf5050

ORGANIC CHEMISTRY

Enantioselective cyanation of benzylic C–H bonds via copper-catalyzed radical relay

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Direct methods for stereoselective functionalization of sp^3 -hybridized carbon–hydrogen $[\text{C}(\text{sp}^3)\text{–H}]$ bonds in complex organic molecules could facilitate much more efficient preparation of therapeutics and agrochemicals. Here, we report a copper-catalyzed radical relay pathway for enantioselective conversion of benzylic C–H bonds into benzylic nitriles. Hydrogen-atom abstraction affords an achiral benzylic radical that undergoes asymmetric $\text{C}(\text{sp}^3)\text{–CN}$ bond formation upon reaction with a chiral copper catalyst. The reactions proceed efficiently at room temperature with the benzylic substrate as limiting reagent, exhibit broad substrate scope with high enantioselectivity (typically 90 to 99% enantiomeric excess), and afford products that are key precursors to important bioactive molecules. Mechanistic studies provide evidence for diffusible organic radicals and highlight the difference between these reactions and C–H oxidations mediated by enzymes and other catalysts that operate via radical rebound pathways.

The clinical success of potential drug candidates has been shown to correlate with the fraction of sp^3 -hybridized carbon atoms and the number of stereogenic centers in the molecule (*1*). Consequently, synthetic methods that enable stereoselective functionalization of $\text{C}(\text{sp}^3)\text{–H}$ bonds could have a major effect on the discovery and development of new pharmaceuticals. Enzymatic oxygenases and halogenases, such as cytochrome P450 and nonheme iron enzymes (*2–4*), have inspired the discovery of synthetic methods for selective oxidation of C–H bonds [see, for example, (*5–8*)]. These enzymes and related small-molecule catalysts function via generation of a reactive metal-oxo species that abstracts an H atom from a $\text{C}(\text{sp}^3)\text{–H}$ bond, followed by radical rebound (*9*) to the metal-bound hydroxide or halide group (Fig. 1A). Here, we report a complementary radical relay strategy for $\text{C}(\text{sp}^3)\text{–H}$ oxidation whereby a reactive radical ($\text{X}\cdot$) abstracts a hydrogen atom from an sp^3 -hybridized carbon atom to produce a diffusible organic radical. This radical is then trapped by a second species that generates the functionalized sp^3 center (Fig. 1B). This concept is demonstrated in copper-catalyzed enantioselective cyanation of benzylic C–H bonds (Fig. 1C). Optically pure alkyl nitriles are featured in many bioactive natural products, pharmaceuticals, and agricultural chemicals (*10*). Moreover,

benzylic nitriles are direct precursors to phenethylamine derivatives, which represent a broad class of hormones, neurotransmitters, and psychoactive drugs (*11*), as well as arylacetic acid derivatives, which are among the most widely used nonsteroidal anti-inflammatory drugs (NSAIDs) (*12*). The stereoselectivity is proposed to arise from reaction of an achiral carbon radical with a chiral copper(II) species ($\text{L}^*\text{Cu}^{\text{II}}\text{–CN}$) (*13*), a pathway that contrasts with most enantioselective C–H functionalization reactions that involve selective, stereospecific activation of a prochiral C–H bond within an organic molecule (*14, 15*).

In recent years, we have been investigating Cu-catalyzed methods for oxidation of styrenes and other alkenes [for example, (*16, 17*)], as well as C–H bonds (*18*). The former reactions are proposed to proceed via reaction of a benzylic radical intermediate with copper(II), generating an organocopper(III) species that undergoes reductive elimination to afford a new C–C or C–heteroatom bond. This pathway, which resembles a number of recently reported nickel- and copper-catalyzed cross-coupling methods that involve the reaction of a carbon-centered radical with a transition-metal center [e.g., (*13, 19–23*)], prompted us to consider how analogous concepts could be applied to C–H oxidation. Kharasch-Sosnovsky allylic oxidations and related reactions provide relevant precedent for this idea (*24, 25*); however, these reactions face constraints that have greatly limited their synthetic utility over the nearly 60-year history of the reaction. For example, the alkene substrate is almost always used in large excess relative to the peroxide-based oxidant, a feature that severely restricts the net product yield and, in turn, the scope of suitable substrates. The majority of examples feature only simple cyclic alkene substrates, such

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Science **353** (6303), 1011-1014. [doi: 10.1126/science.aaf5050]

Editor's Summary

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